Econometrics Lecture Notes

EC 325 | Colby College

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Table of contents

Preface

About these Notes

This booklet contains the lecture notes for EC 325: Econometrics at Colby College. These notes are meant to supplement the course readings.

The content in this book is *heavily inspired* by three main texts, which are required for this course:

1. Wooldridge (2019)
2. Angrist and Pischke (2014)
3. Wickham, Çetinkaya-Rundel, and Grolemund (2023)

Structure of These Notes

This course is basically broken into three modules:

1. Correlation, Causation, and Regression
2. Real-Life Regression Models
3. Introduction to Causal Methods

Each chapter contains both notes for the course - in narrative form - as well as econometric labs that we will work through in R.

Roadmap

- In Chapter 1, we will discuss the difference between correlation and causality, and what it means for us as economists.

Statistical Software

Almost all modern econometrics is done with specialized statistical software which you interact with via code.

About Me

I am a recent PhD from UMass Amherst. My research focuses on the relationship between public/economic policy and maternal, infant, and child health and racial/ethnic health disparities.

1 Introduction

This is a book created from markdown and executable code.

Key Concepts

- Learn the difference between correlation and causation.

1.1 What is this Class About?

This class is obviously dedicated to the study of econometrics (endearingly shortened 'metrics). But what exactly does econometrics entail? What differentiates it from broader terms such as "statistical analysis" or "data analysis"?

From Wooldridge (2019), we get a classic definition of econometrics:

> "…statistical methods for estimating economic relationships, testing economic theories, and evaluating and implementing government and business policy."

However, my view of econometrics is at the same time, broader and more narrow:

Econometrics

The practice of using statistical, quantitative methods to study social phenomena and provide insights into public policy. More specifically, modern econometrics has focused on identifying the **causal effects** of social phenomena/policy as opposed to **correlations**.

In other words, I view the practice of econometrics as the use of quantitative methods to understand the social world around us, with a special (though not exclusive) lens focusing on separating correlation from causation.

1.2 Does Having Health Insurance *Cause* You to be Healthier?

Let's think about a critical question in public policy: *does having health insurance make someone healthier*? This is especially an important question in the context of the United States, which does not have a public health insurance system. Instead, it is a predominantly private system entangled with a complex net of social safety nets meant to provide a patchwork of coverage to the most marginalized.

However, as of XXX, YYY do not have any health insurance coverage. Would the US as a whole be better off guarantee health insurance coverage for everyone? If health insurance makes people healthier, then there may be a strong case to do so. If not, the benefits of such a system would become, at the very least, more opaque.

2 Summary

In summary, this book has no content whatsoever.

References

Angrist, Joshua D., and Jörn-Steffen Pischke. 2014. *Mastering 'Metrics: The Path from Cause to Effect*. Princeton University Press.

Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O'Reilly Media, Incorporated.

Wooldridge, Jeffrey M. 2019. *Introductory Econometrics: A Modern Approach*. 7th ed. Cengage Learning.